

Photon Transfer Curve — pco.edge 10 bi CLHS Noise Decomposition

Setup

Following the same two-frame-difference method as the ThorCam analysis ([photon_transfer.md](#)), we recorded bright/dark frame stacks for the pco.edge 10 bi CLHS (serial 22500190) across 38 illumination levels: a 30-level base sweep spanning 0.1%–99.9% of full well (log-spaced target fractions), plus an 8-level extension densely filling 80%–99% of full well. For each level: 50 bright frames (filter wheel open) and 50 dark frames (filter wheel closed, back-to-back), both restricted in hardware to the central 10% ROI and further cropped to the central 70×70 px sub-ROI in analysis (the same ROI size validated vignetting-free for the ThorCam). Exposure at each level was set by autoexposure, ranging 0.007–1897 ms. This camera has no gain control — the `gain_range` field in the recorded data is a layout placeholder, not a real setting. The camera was configured in **fast scan mode** — this is a camera-level readout setting made outside this codebase (`instruments/pypcocam/pcocam.py` exposes no scan-mode setter, and it is not recorded in the saved `.npz` metadata), so it is stated here rather than derived from the data. Fast scan trades read noise for readout speed relative to the camera's slower "normal"/low-noise scan modes; see the read-noise caveat below. Acquisition: [scripts/test/pco_noise_bright.py](#). All figures and numbers below are reproduced from [pco_noise_analysis.ipynb](#).

Parameter	Value
Serial	22500190
Signal levels	38 (30 base, 0.1%–99.9%; 8 extension, 80%–99%, densely sampled)
Frames per level	50 (bright), 50 (dark)
ROI	central 70×70 px (of a central-10%-of-sensor hardware ROI)
ADC full-scale (<code>pixel_max</code>)	65,535 DN (16-bit)
Exposure range	0.007–1897 ms (autoexposure)
Readout mode	fast scan
Gain control	none

Parameter	Value
Dark subtraction	master dark (mean of 50 dark frames per level)

The base and extension acquisitions overlap at the top end (base's last level reaches 64,493 DN at 1852 ms; the extension spans 51,965–64,920 DN over 1486–1897 ms) and the extension frames are 8 px wider than the base frames (448 vs. 440 px) — both stacks are concatenated, the extension is cropped symmetrically to match the base ROI center, and the combined levels are sorted by exposure before analysis so every vs.-exposure plot reads monotonically. The three shortest exposures are pinned at the 7 μ s hardware minimum and sit at only ~ 0.1 – 0.5 DN of *light* signal above the dark pedestal — the left end of the curve, not a measurement error.

Noise Model

Identical model to the ThorCam analysis — see [photon_transfer.md](#) , ["Noise Model"](#) for the full derivation. In short:

$$\text{sigma_total}^2 = \text{sigma_read}^2 + \text{sigma_shot}^2 + \text{sigma_fpn}^2$$

with `sigma_read` the signal-independent read-noise floor, `sigma_shot` = $\sqrt{S/K}$ the photon shot noise, and `sigma_fpn` = $P_N * S$ the fixed-pattern noise. The two-frame difference method ($\text{sigma_temporal}^2 = \text{var}(F1 - F2)/2$, mean-subtracted per pair) isolates `sigma_read` + `sigma_shot` from FPN and from lamp flicker; `sigma_fpn` is then recovered by quadrature subtraction from the single-frame spatial std, `sigma_total` .

Conversion Gain and Read Noise

K is fit from $\text{sigma_temporal}^2 = \text{sigma_read_var} + S/K$, restricted to the linear shot region (1%–60% of full well). Read noise is measured directly from the dark-frame stacks (two-frame difference method), not extrapolated from the PTC intercept.

Photon transfer curve -- pco.edge 10 bi CLHS

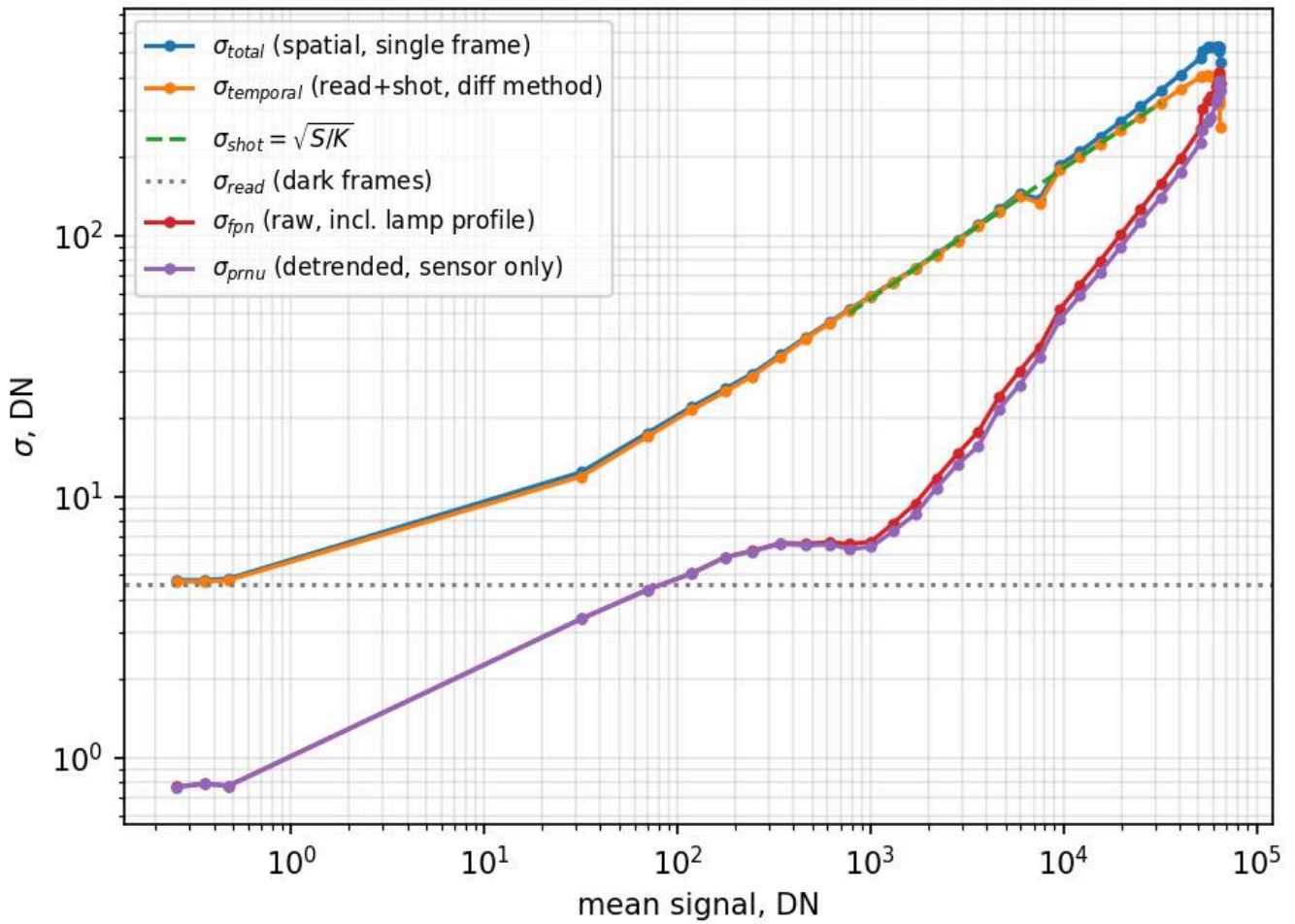


Figure 1. Full noise decomposition: σ_{total} (spatial), σ_{temporal} (difference method), σ_{shot} (fit), σ_{read} (dark-frame floor), σ_{fpn} (raw, quadrature), σ_{prnu} (detrended, quadrature). Log-log.

Quantity	Value	Source
Conversion gain, K	0.308 e ⁻ /DN	fit (linear region)
Read noise (direct), upper bound	4.614 DN (1.420 e ⁻)	dark-frame difference, this dataset
Full well (<code>pixel_max * K</code>)	≈20,200 e ⁻	ADC-clip upper bound
Dynamic range (<code>pixel_max / read noise</code>)	≈14,200:1 (~83 dB), conservative	quantization-limited

Unlike the ThorCam, there is no dedicated `pco_ptc_gain.py` / `pco_read_noise.py` sweep for an independent cross-check of K and read noise yet — the values above come from this single dataset only.

Read noise is an upper bound, not a clean measurement, for this dataset. The lamp was bright enough that autoexposure hit the camera's 7 μ s hardware exposure minimum before

reaching the dimmest target signal levels — the three shortest-exposure levels are all pinned at 7 μ s (two of them) and 14 μ s rather than continuing to shrink with target fraction. This dataset was never designed to isolate read noise the way a dedicated minimum-exposure dark sweep would, so it doesn't cleanly reach the read-noise floor. Compounding this, the dark-frame *mean* at the short-exposure end **decreases** with exposure (101.03 $\rightarrow$ 100.82 $\rightarrow$ 100.69 $\rightarrow$ 100.34 DN over 7 μ s $\rightarrow$ 7 μ s $\rightarrow$ 14 μ s $\rightarrow$ 1076 μ s) — the opposite of what dark current would produce — so the minimum-exposure dark frames are not behaving as a clean, stable read-noise reference either. `sigma_read` = 4.614 DN is therefore reported as an upper bound; a dedicated read-noise sweep (the pco analogue of [scripts/test/thorcam_read_noise.py](#), which doesn't exist yet for this camera) is needed for a real number, and would also let a lower-noise scan mode be checked against fast scan mode (Setup, above).

Fixed-Pattern Noise: Raw vs. Detrended

As with the ThorCam, the raw `sigma_fpn` includes both sensor PRNU and the lamp's own spatial illumination profile across the ROI; subtracting a low-order 2D polynomial (fit per level from the mean image) isolates true pixel-to-pixel PRNU as `sigma_prnu`.

Quantity	Value
PRNU factor, <code>P_N</code> (detrended, sensor-only)	$\approx$ 0.44% of signal
Raw FPN factor (includes lamp profile)	$\approx$ 0.50% of signal

The gap between raw and detrended is narrower here than for the ThorCam (0.44% vs. 0.50%, compared to 0.53% vs. 0.60%) — for this camera the raw curve is nearly indistinguishable from PRNU except very close to full well (Figure 1, upper right), where the lamp/vignetting gradient becomes visible in both cameras' data.

Lamp Stability Check

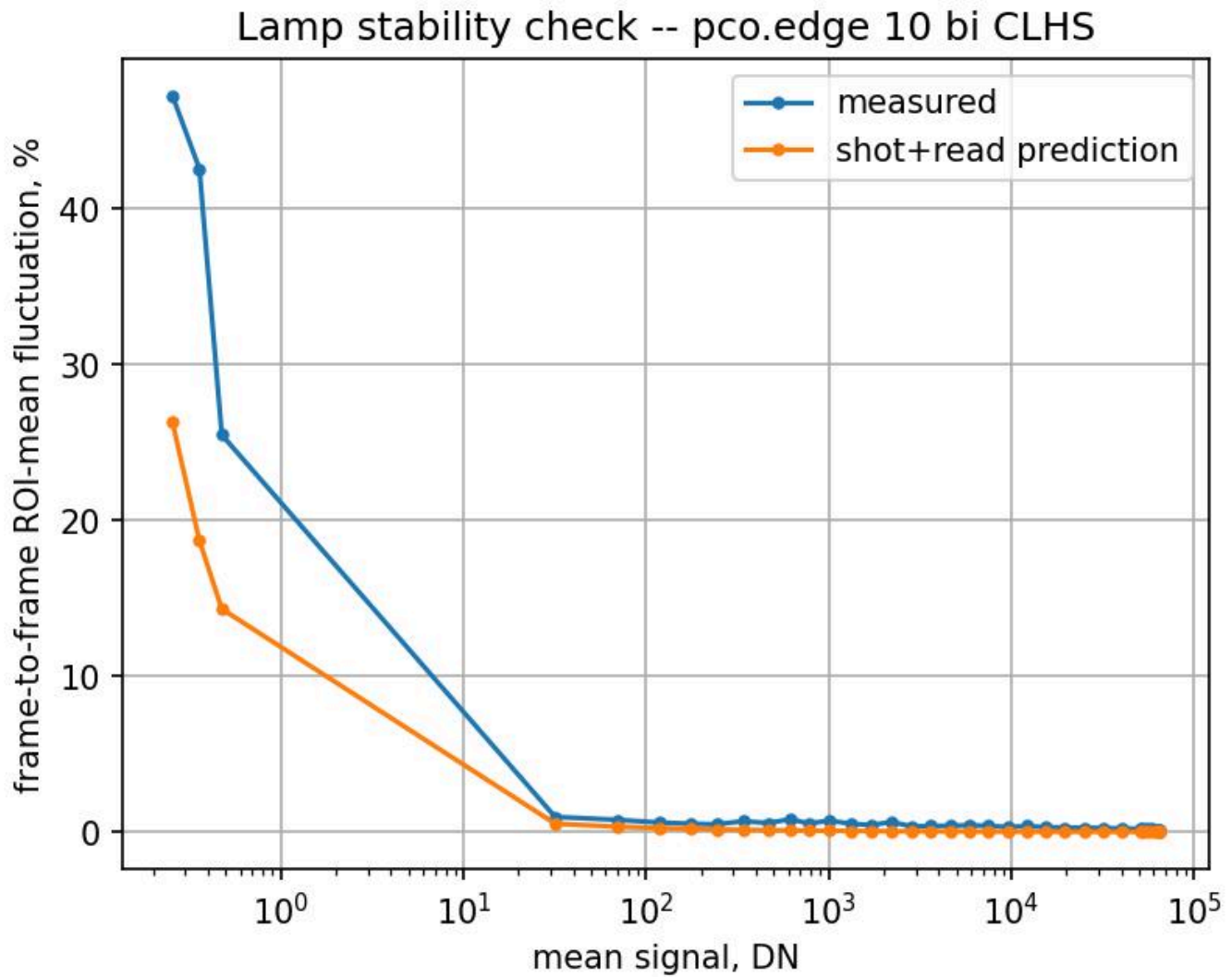


Figure 2. Measured frame-to-frame ROI-mean fluctuation (%) vs. the shot+read-noise prediction, across signal levels.

The measured curve sits above the shot-noise prediction at every level, by $\sim 0.07\text{--}0.21\%$ at high signal (and much more at very low signal, where the percentage is dominated by the read floor and the ROI mean is only a few DN above the dark pedestal). As with the ThorCam, this lamp flicker does not corrupt `sigma_temporal` in Figure 1, because mean-subtracting each difference image removes exactly this common-mode fluctuation before the variance is taken.

Camera Comparison: pco.edge 10 bi CLHS vs. ThorCam CS126

Both cameras were run under the same lamp and analysis pipeline (`noise_decomposition()` in `pco_noise_analysis.ipynb`, applied identically to both), so the numbers below are directly comparable. DN values are *not* comparable between the two cameras — K differs by a factor of ~ 8.6 (0.308 vs. 2.660 e^-/DN) — so all noise comparisons below are converted to electrons.

Quantity	pco.edge 10 bi CLHS	ThorCam CS126
Conversion gain, K (e ⁻ /DN)	0.3078	2.6596
Signal rate at matched exposure (e ⁻ /ms)	10.74	6.31
Read noise (DN), pco is an upper bound	4.614	0.275
Read noise (e ⁻), pco is an upper bound	1.420	0.731
Full well (e ⁻)	≈20,200	≈10,900
Dynamic range (dB), pco is conservative	≈83.0	≈83.5
PRNU factor, P _N	≈0.44%	≈0.51%
ADC	16-bit (65,535 DN)	12-bit (4,095 DN)
Readout mode	fast scan	(see photon_transfer.md)
ROI / frame size	236×440–448 px	300×408 px

The pco's read-noise and dynamic-range figures carry the same upper-bound caveat as in the "Conversion Gain and Read Noise" section above — the pco values in this table may overstate its read noise (understate its dynamic range) relative to a dedicated measurement.

(ThorCam figures in this table come from a 20-level subset used specifically for this comparison and agree with the full 26-level sweep in `photon_transfer.md` to <3%.)

Signal rate and sensitivity

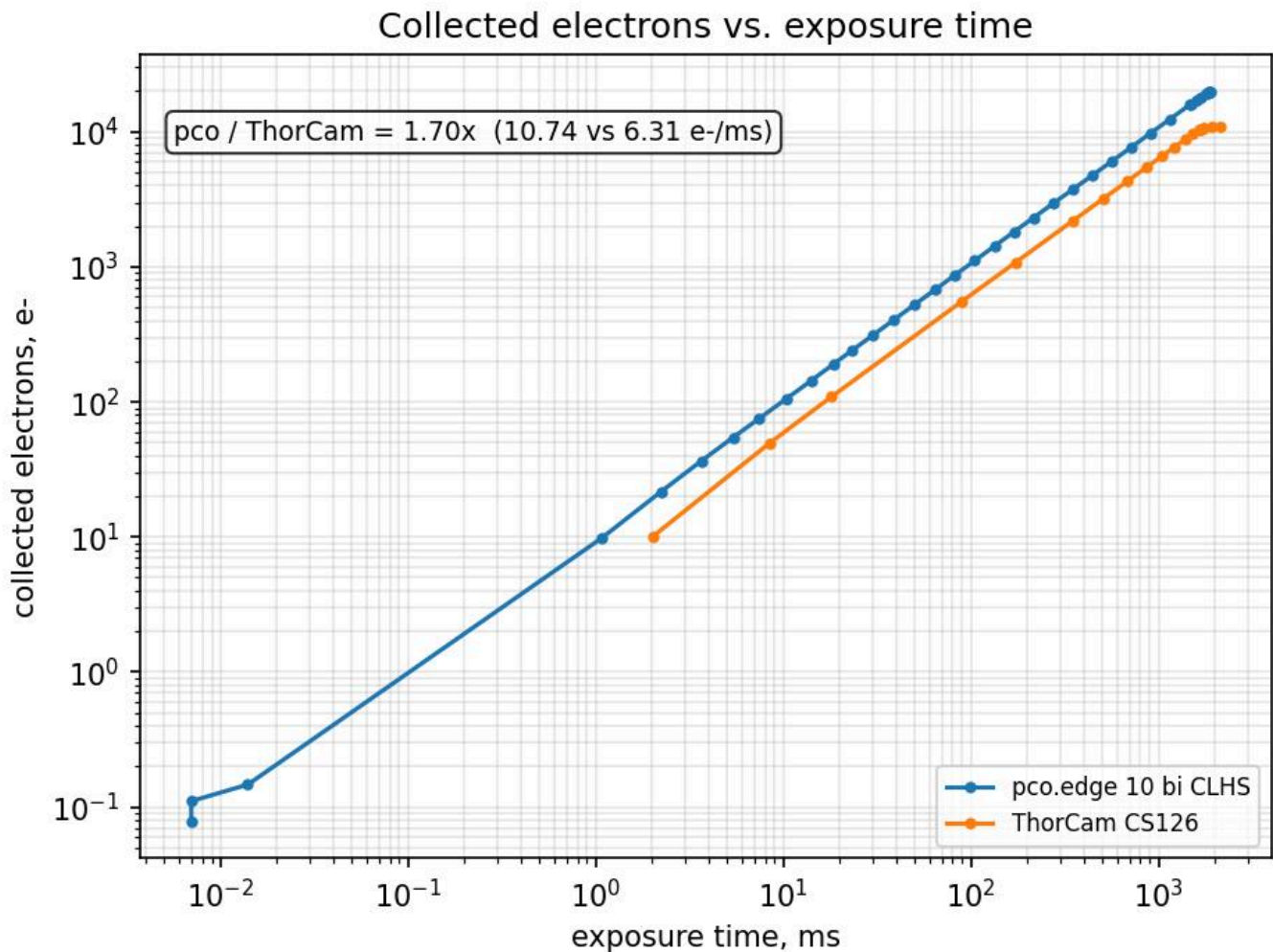


Figure 3. Collected electrons (mean signal $\times$ each camera's own K) vs. exposure time, under identical lamp illumination. Annotated with the headline ratio.

At the same exposure time, the pco collects 1.70 $\times$ more photoelectrons than the ThorCam

(10.74 vs. 6.31 e-/ms — an origin-forced linear fit of electrons vs. exposure time, each camera restricted to its own 1%–60%-of-full-well linear region). This is a single, flat ratio: computed pointwise over the exposure range where both cameras are in their linear region (23–860 ms), it stays within 1.70–1.72 $\times$ at every point. The advantage reflects some combination of pixel size, quantum efficiency, and the different ROI/optical coupling between the two acquisitions; with $\sim 1.9\times$ the full well on top of the faster collection rate, the pco also keeps accumulating for longer before saturating.

Relative error vs. exposure time

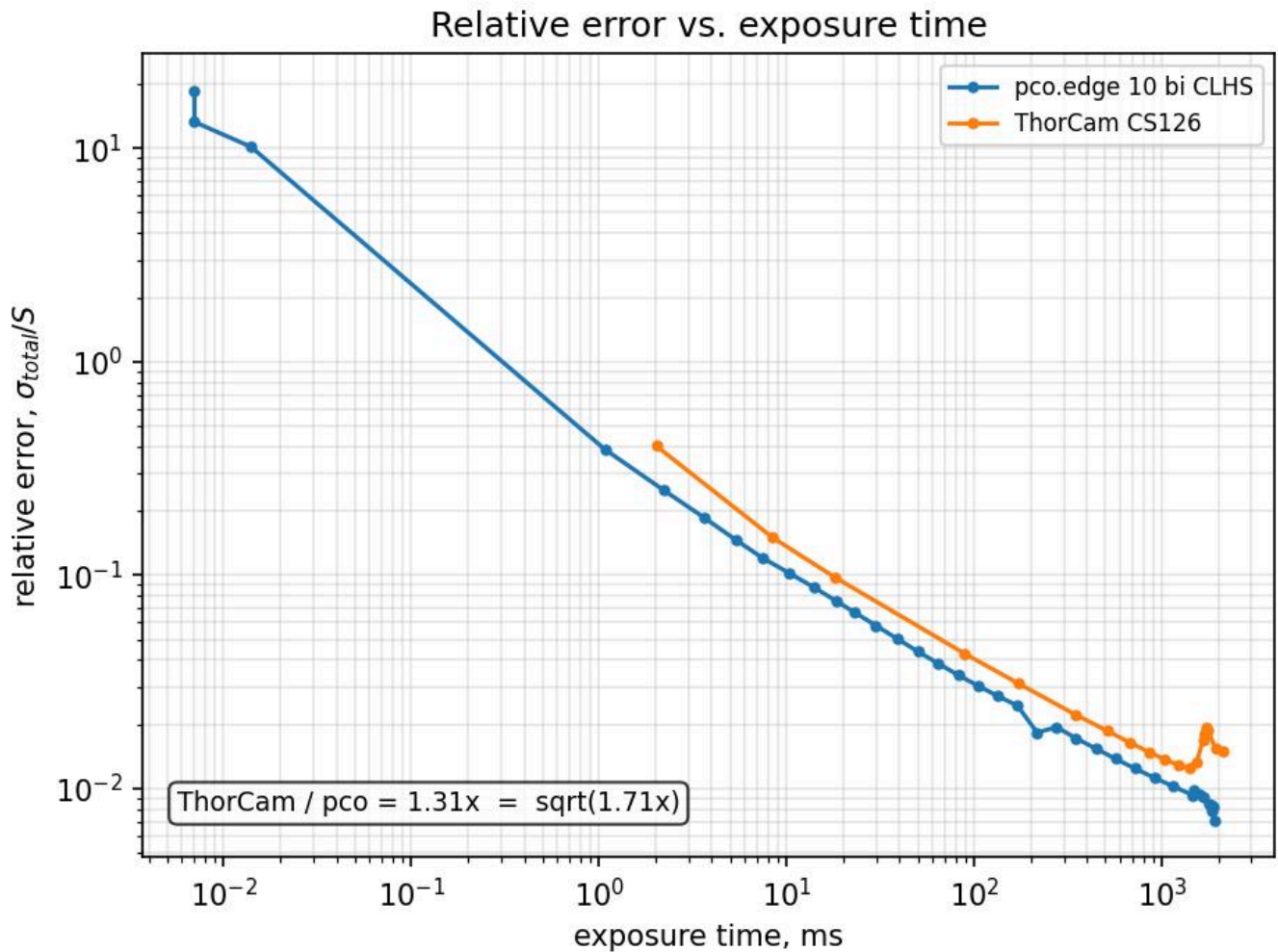


Figure 4. Single-frame relative error (σ_{total} / S) vs. exposure time, both cameras. Annotated with the headline ratio.

At the same exposure time, the pco's relative error is 1.31× lower than the ThorCam's (pointwise over the same 23–860 ms linear-region overlap, range 1.28–1.45×). This is not an independent effect: $\sqrt{1.708} = 1.307$ — the accuracy advantage is exactly the shot-noise consequence of the signal-rate advantage in Figure 3. Per collected electron, the two cameras are equally accurate; the pco is simply ahead because it collects electrons faster. Both curves converge toward the same PRNU-limited floor (~0.4–0.5%) at long exposure, where fixed pattern noise rather than shot noise dominates.

Noise in electrons

Noise comparison in electrons: pco.edge 10 bi CLHS vs ThorCam CS126

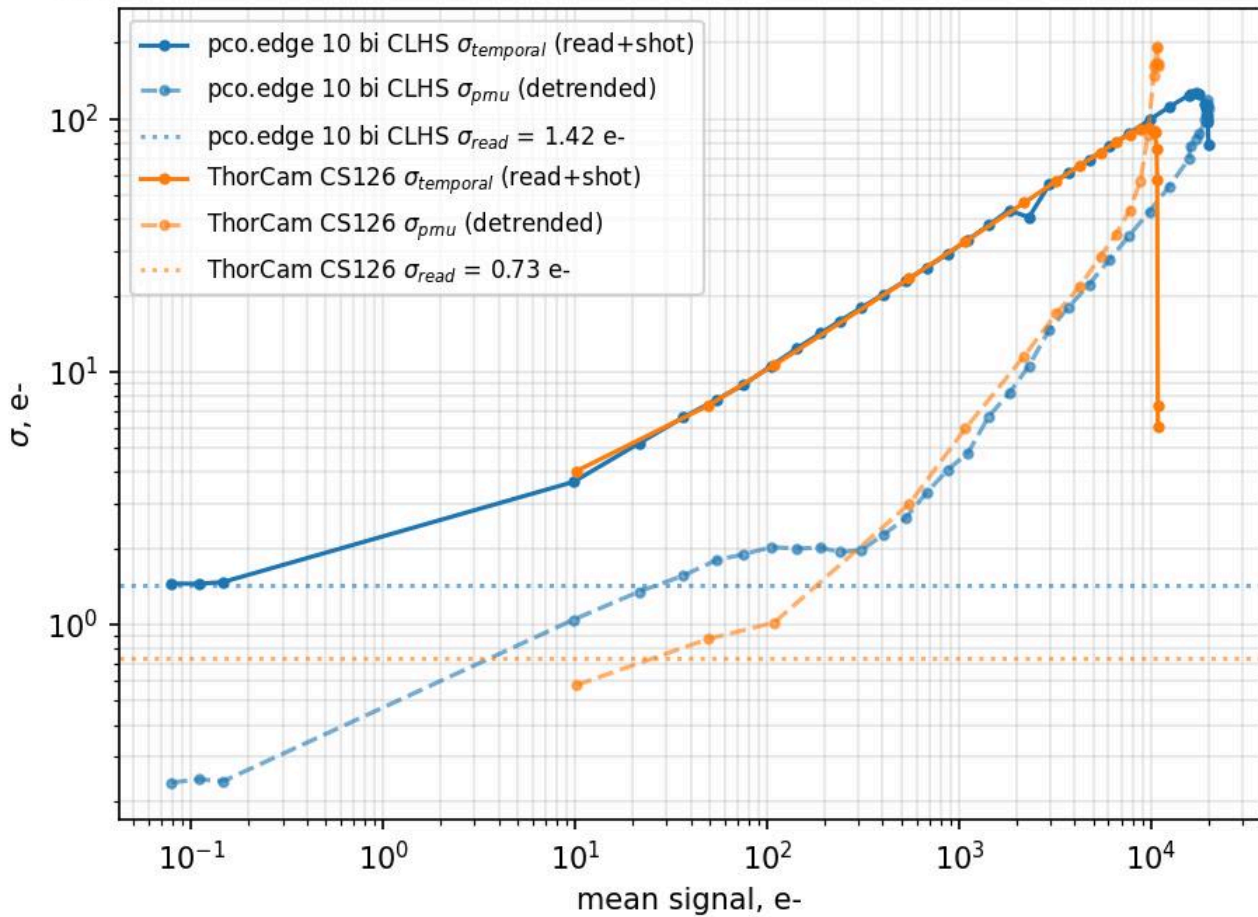


Figure 5. σ_{temporal} (read+shot) and σ_{prnu} (detrended), both in electrons, vs. mean signal in electrons, for both cameras, with each camera's electron-referred read-noise floor as a dashed horizontal line.

In electrons, the pco's read-noise floor (1.42 e⁻, an **upper bound** — see the caveat above) sits about 2× above the ThorCam's (0.73 e⁻) — on this dataset the pco appears noisier at low signal, though a dedicated read-noise sweep could narrow that gap. The two σ_{temporal} curves converge onto the same shot-noise slope once the read floor is left behind, and the two σ_{prnu} curves track closely across the shared signal range, consistent with the similar P_N values in the table above. The apparent spike in the ThorCam σ_{temporal} curve near its top signal levels is sampling noise from that dataset's few near-saturation levels, not a measurement artifact.

Which camera for which measurement

- **At matched exposure time**, the pco reaches a **1.31× lower relative error** than the ThorCam across the shared linear-region range (Figure 4) — the shot-noise consequence of collecting **1.70× more electrons** per unit exposure (Figure 3). Combined with ~1.9× the

full well, it also saturates later, extending this advantage to longer exposures / brighter scenes.

- **At matched signal (electron count)** rather than matched exposure time — e.g. a frame-rate-limited application where exposure can't simply be extended — the ThorCam's lower read-noise floor (0.73 vs. an **upper-bound** 1.42 e⁻ for the pco, Figure 5) would give it the edge in the read-noise-dominated regime, at very low electron counts; a dedicated pco read-noise sweep is needed to say by how much.
- **PRNU-limited accuracy** (long averaging, flat-fielded or ratio measurements) is essentially a wash — both cameras plateau at a similar ~0.4–0.5% PRNU floor.

Conclusions

- Both cameras follow the same Janesick noise model; the main practical difference is the ~8.6× larger conversion gain step (fewer, noisier electrons per DN) and (at most, see the read-noise caveat below) ~2× larger read noise in electrons for the pco, traded against ~1.9× more full well and a measured **1.70×** faster electron-collection rate at matched exposure time — which alone accounts for the pco's **1.31×** lower relative error at matched exposure ($\sqrt{1.70} = 1.31$).
- Lamp noise is real but secondary for the pco, as it was for the ThorCam: ~0.1–0.2% temporal flicker (removed by mean-subtracting each difference pair) and a small spatial illumination gradient across the ROI (removed by polynomial detrending before computing PRNU).

Derived constants (this camera, serial 22500190 , fast scan mode, no gain control):

Constant	Value
Conversion gain, K	0.308 e⁻/DN
Read noise, sigma_read — upper bound	4.614 DN ≈ 1.420 e⁻
Full well	≈20,200 e⁻ (pixel_max × K , ADC-clip bound)
Dynamic range — conservative (uses the upper-bound read noise)	≈14,200:1 (~83 dB)
PRNU factor, P_N (sensor only)	≈0.44% of signal
Raw FPN factor (incl. lamp profile)	≈0.50% of signal
Lamp temporal instability (excess)	≈0.07–0.21% frame-to-frame at high signal
Signal rate vs. ThorCam CS126, at matched exposure	1.70× more e ⁻ /ms

Constant	Value
Relative-error ratio vs. ThorCam CS126, at matched exposure	1.31× lower

Fast Error Estimation

Same estimators as [photon_transfer.md](#), re-derived with this camera's constants. S is mean signal in DN, N is the number of frames averaged. Because σ_{read} is an upper bound (see the read-noise caveat above), eps_int below is **conservative** — it will overstate the true error somewhat at low signal, where the read term matters most. The eps_T transmission-ratio formulas and the quick-reference table are shot-noise-dominated in the signal range this camera is normally used at and are effectively unaffected.

Per-pixel intensity, single ROI

$$\text{eps_int}(S, N) \text{ [\%]} = 100 * \sqrt{(21.29 + 3.249*S)/N + (0.00442*S)^2} / S$$

- $21.29 = \sigma_{\text{read}}^2$ (**upper bound** — see the read-noise caveat above), $3.249 = 1/K$ (DN^2/DN , shot term), $0.00442 = P_N$ (PRNU).
- Shot-limited quick form (valid above a few hundred DN, single frame):
 $\text{eps_int} \approx 180.3/\sqrt{S} \text{ \%}$.
- **Averaging floor:** the $0.00442*S$ (PRNU) term does not shrink with N . Beyond a few hundred frames, per-pixel intensity error stalls at **≈0.44%**, no matter how long you average. Only flat-fielding / referencing removes it (next section).

Transmission ratio (the metasurface case)

For $T = S_{\text{sample}} / S_{\text{ref}}$, the fixed per-pixel PRNU term cancels in the ratio and the error is purely temporal (shot+read):

$$\text{eps_T}(S_s, N_s, S_r, N_r) \text{ [\%]} = 100 * \sqrt{1/(K*S_s*N_s) + 1/(K*S_r*N_r)}$$

Dim/shot-limited single-term approximation (reference much brighter/better averaged than the sample, so the sample term dominates):

$$\begin{aligned} \text{eps_T} \text{ [\%]} &\approx 180.3 / \sqrt{S_s * N_s} & (S_s = \\ &\text{sample/transmitted signal, DN}) \\ N_s \text{ (for target eps\%)} &\approx 32,494 / (S_s * \text{eps}^2) \end{aligned}$$

not average down

$$S_{\text{electrons}} = S_{\text{DN}} * K$$